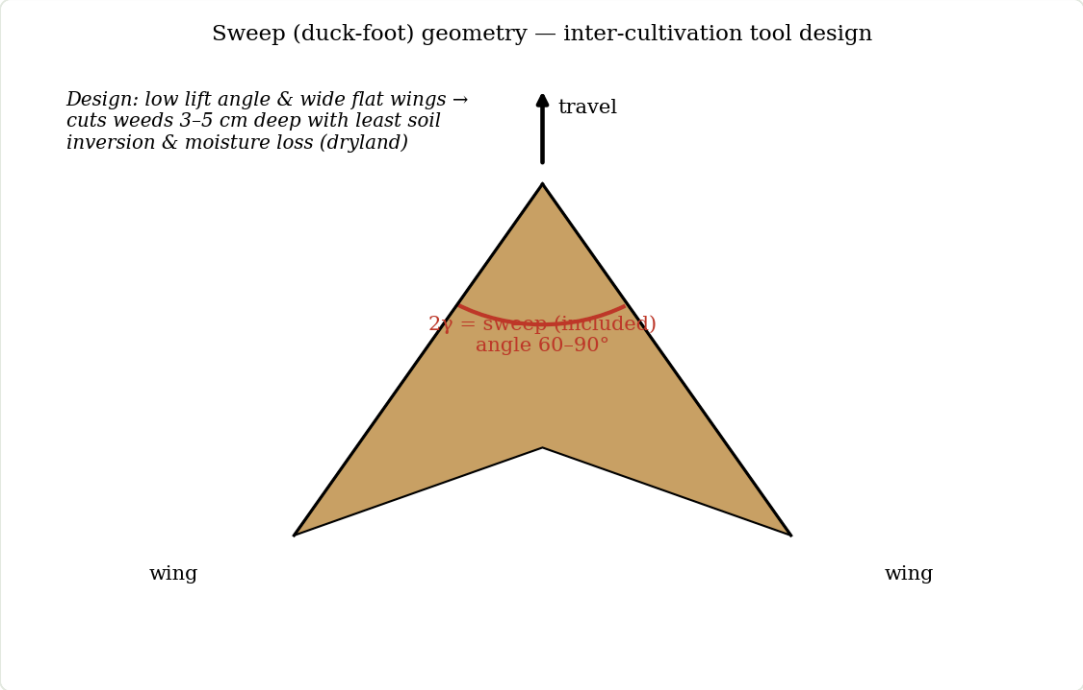
UNIT 2 · TOPIC 7(B)

Design of Inter-Cultivation Equipment

Syllabus topic: Design of fertilizer applicators and inter-cultivation equipment

Primary purpose: weed control. Moisture-mulch and aeration benefits are secondary — Kepner cites weed control as the chief benefit of cultivation (Ch. 12).

Sweep (duck-foot) design



Sweep geometry

- Included (sweep) angle **60–90°**; a low lift/pitch angle so weeds are cut and soil is not inverted; working depth **3–5 cm** near rows; wing widths overlap for full cutting.
- Speed raises soil throw — shields/fenders protect young plants. The minimum untilled band on each side of the row shrinks as plants age.

Shanks (standards)

Shank	Features
Rigid	With shear pin
Spring-trip	Auto-reset; set the trip force above working resistance (~1.5–2× draft)
Spring (vibrating)	Shallow work
C-shank vs S-tine	Differ in stiffness

Depth and alignment

Gauge wheels and parallel-linkage gangs keep sweep pitch constant over undulations — a key design feature of modern cultivators. Use front-mounted or rear tool-bar mounting for visibility.

Rotary tools

- **Rotary hoe:** rigid spider wheels with curved teeth, ground-driven; works at **8–16 km/h**. The teeth enter vertically and lift out backward, uprooting small weeds ("blind cultivation" pre/post-emergence) with minimal crop damage.
- **Rotary (power) weeder:** L-blades on a horizontal shaft; blade tip speed and bite length set weed cutting vs power (same rotor mechanics as the rotavator); shields limit throw onto rows.

Design draft values

- Row-crop cultivator ≈ **30–80 kgf per sweep** (shallow).
- Wheel hoe manual force ≈ **8–12 kgf** sustained (ergonomic limit ~**100–150 W** human power — Unit 5).

Practice MCQs

Q135. According to Kepner, the chief benefit of row-crop cultivation is:

(a) Aeration
(b) Moisture mulch
(c) Weed control
(d) Root pruning

Q136. The included angle of cultivator sweeps is generally:

(a) 20–30°
(b) 60–90°
(c) 100–120°
(d) 150°

Q137. Sweeps for dryland moisture conservation are designed with:

(a) High lift and pitch
(b) Low pitch/lift — cut weeds without inversion
(c) Deep suction
(d) Narrow wings

Q138. Near young row crops, cultivation depth should be:

(a) 3–5 cm
(b) 10–12 cm
(c) 15–20 cm
(d) 25 cm

Q139. Spring-trip cultivator standards are set to trip at about:

(a) Working draft
(b) 1.5–2× working draft, with auto-reset
(c) 5× draft
(d) Any load

Q140. Shields on row-crop cultivators:

(a) Reduce draft
(b) Protect young plants from soil throw
(c) Meter seed
(d) Firm soil

Q141. The parallel-linkage gang mounting on cultivators keeps:

(a) Speed constant
(b) Sweep pitch/depth constant over uneven ground
(c) Row spacing fixed
(d) Draft zero

Q142. The rotary hoe is used for:

(a) Deep tillage
(b) Blind (pre/early post-emergence) shallow cultivation at high speed
(c) Puddling
(d) Ridging

Answers

Q135 — (c) Weed control
Q136 — (b) 60–90°
Q137 — (b) Low pitch/lift — cut weeds without inversion
Q138 — (a) 3–5 cm

Q139 — (b) 1.5–2× working draft, with auto-reset

Q140 — (b) Protect young plants from soil throw

Q141 — (b) Sweep pitch/depth constant over uneven ground

Q142 — (b) Blind (pre/early post-emergence) shallow cultivation at high speed

Sources: Kepner, Bainer & Barger — Principles of Farm Machinery, Ch. 11 (Crop Planting), Ch. 12 (Row-Crop Cultivation), Ch. 13 (Application of Fertilizers); standard FMP design references.
